

Project 1 on Machine Learning: regression analysis, resampling methods and gradient descent

Tim Herrmann Holmen

FYS-STK4155

Department of Physics

Dette prosjektet er en vandring gjennom metodene innen dataanalyse som har ledet til det vi idag kaller maskinl ring. Hovedm let er   studere i detalj diverse regresjonsmetoder, som ordinary least squares, ridge regresjon og lasso regresjon. Resampling teknikker som bootstrap og kryssvalidering lar oss vurdere og velge den beste metoden for problemet. For   optimalisere parametrene (hvilke parametre er det her snakk om?: det er selve parametrene og hyperparametrene) vil vi bruke gradient descent, og teste ulike metoder for   utf re denne beregningen, som best r av derivasjon. Vi har studert newtons metode og automatiserte derivasjons metoder, for   optimalisere parametrene i modellene. Polynomene vi utvikler skal passe til en spesifikk en dimensjonal funksjon kalt Runges funksjon, med et st y element, dermed har vi det analytiske svaret tilgjengelig.

CONTENTS

I. Introduction	1
A. 2a The motivation.	1
B. 2b What has been done and what i do.	2
1. Linear Models	2
2. Resampling techniques	2
3. Optimization	2
4. Cross-Validation	2
C. The structure of the report,	2
II. Methods	2
A. The data	2
III. Conclusion	2
References	2

I. INTRODUCTION

A. 2a The motivation.

The reason we want to fit a function to data is that we want to predict the value of the function to new unknown data points. For fitting a function to data points, we use polynomials, and we want to find the best polynomial that fits the data. The best polynomial is the one that minimizes the mean squared error (MSE) between the predicted values and the true values (for the known data). The MSE can be decomposed into three components: bias, variance and irreducible error.

$$MSE = Bias^2 + Var[\tilde{y}] + irreducible\ noise$$

The bias measures how far the average prediction is from the truth. The variance measures how much the prediction moves when the training set is redrawn. The irreducible error is the variance of the noise itself, which is baked into the data and the model cannot predict it. (Hjorth-Jensen, 2016, p. 62) The bias-variance tradeoff is the balance between bias and variance, in which we

minimize the MSE. We do not want an overfitted model, which has low bias and high variance, nor do we want an underfitted model, which has high bias and low variance. Optimally, we want a model that is just right, with low bias and low variance. When fitting a polynomial to data, we want to find the right degree of the polynomial and the right parameters for each degree. For example if there is a linear relationship between the data points, and we can assume this by looking at the data, we use a second order polynomial to describe the data, and the parameters are the coefficients of the polynomial, which can be changed in accordance with the relationship between the data points. The parameters are found by minimizing the MSE between the model prediction and datapoints, for avoiding overfitting, we can split the data into training and testing sets, to see how well the model performs on unseen data. We can use regularization methods, such as ridge regression and lasso regression, to regularize the model and avoid overfitting, and thus lower the variance, while maintaining a good bias. We can also use resampling methods, such as bootstrap and k-fold cross validation, to estimate the performance of the model on unseen data, and thus choose the best model for the data.

With lasso regression, we do not have the analytical solution for the gradient, as we do for OLS and Ridge, and we have to use gradient descent to find the optimal parameters. "Gradient descent is an optimization algorithm inspired by newtons method, which is used to find the minimum of a function. The function we want to minimize is the cost function, which is the MSE between the predicted values and the true values. The gradient descent algorithm uses the derivative of the cost function to find the direction of steepest descent, and then takes a step in that direction, with a step size determined by the learning rate. We can use different optimization algorithms, such as momentum, adagrad, rmsprop and adam, where the learning rate is dependent on the iteration number to improve the convergence of gradient

descent.” - AI After exploring the different methods, we can compare them with cross validation, and choose the best model for the data.

B. 2b What has been done and what i do.

1. Linear Models

a. *OLS*

b. *Ridge*

c. *Lasso*

2. Resampling techniques

a. *Bootstrap*

b. *Kfold cross validation*

3. Optimization

a. *Gradient descent*

b. *Momentum*

c. *Adagrad*

d. *RMSprop*

e. *Adam*

f. *Stochastic gradient descent*

4. Cross-Validation

C. The structure of the report,

The last paragraph, one sentence per section, and where code can be found. Code can be found on the following Github repository: (Tim, 2026)

II. METHODS

A. The data

In this project, the input data is normally distributed, and the output data is generated by Runge's function, with added noise, Runge Function which is a one-dimensional function of second order.

$$f(x) = \frac{1}{1 + 25x^2}$$

From the beginning, we have created a relation between the input and output data. This is helpful since the project is research in machine learning methods, it is usefull to know the exact solution. The noise is normally distributed, with a mean of 0 and a variable standard deviation. From the form of the runge function, we can see that the function has a maximum at $x=0$, and the function is symmetric around this point, for values larger than $[-1,1]$, the function approaches zero. Thus we will focus on data which is normally distributed between -1 and 1.

III. CONCLUSION

REFERENCES

- (2026), “FYS-STK4155: Prosjektarbeid i faget FYS-STK4155: Anvendt dataanalyse og maskinl ring,” <https://github.com/Timholmen96/FYS-STK4155>.
M. Hjorth-Jensen (2016), *Machine Learning and the Physical Sciences* (Springer).